INCH-POUND

MIL-C-48156 (AR)
AMENDMENT 6
19 September 1996
SUPERSEDING
AMENDMENT 5
04 February 1993

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, XM299
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-48156 (PA), dated 01 September 1974, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, Add new military specification as follows:

"MIL-P-60312 - Parts, Molded, Plastic Foam, Polystyrene (For use with ammunition)"

Delete under Federal Specifications: "L-T-90 - Tape, Pressure Sensitive"

PAGE 3

3.7, Delete paragraph in its entirety.

PAGE 5

* 4.4.1.e, Change to read as follows:

"Heat shrinkable tubing from not more than manufacturer or manufacturer part number."

PAGE 6

4.3.2.1, Critical 1, change "tape" to "heat shrinkable tubing."

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PAGE 7

4.3.2.2, Delete Majors 132, 133 in their entirety.

* Delete Major 134 in its entirety.

Major 135, change to read as follows:

"Tubing missing, loose, wrinkled, separating or inadequately covering external surface ...0.40% ...3.2 ...Visual/manual."

PAGE 8

4.3.2.3, Add Critical 3 which reads:

"3. Firing plug above flush with head surface ...100% ...3.2 ... Gage"

* Add Critical 4 as follows:

"4. Propellant Weight (after loading) ...100%...3.2 ...4.4.3.2.1/4.5.4.1"

* Major 101, Delete in its entirety.

Add new Major and note a. at bottom of page as follows:

"Major 104 Environmental conditions inadequate .../a ...3.2 ...Gage

a. Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected."

* Major 132: Delete in its entirety and substitute:

"132 Percussion primer above flush (max) of body ...0.40% ...3.2...Gage"

Major B, Add new major:

"137 Stakes missing ...0.40%...3.2...Visual"

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* 4.3.2.4, Add Critical 1 as follows:

"1 ...Propellant Weight ...(B)...3.2...4.4.3.2.2/4.5.4.2"

Major 101: Add new defect which reads as follows:

"101 Environmental conditions inadequate .../a...3.2 ...Gage"

* Major 102: Add new defect which reads as follows:

"102 ...Propellant Weight ...(B)...3.2...4.4.3.2.2/4.5.4.2"

Major B, 133: Delete in its entirety.

Major 135: Delete in its entirety.

Major 136: Delete in its entirety and substitute:

"136 Position of outside diameter to major diameter of threads ...0.40% ...3.2 ...Gage"

Note a: Add note a. which reads as follows: "a. Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected."

* Add Note b as follows:

"(b) See 4.4.3.2.2 for defect description. Same sample for both inspections."

PAGE 11

4.3.2.5, Add new major 101 and notes (a) (b) as follows:

"101 Moisture content excessive (a) (b) 5.1 (a) (b)"

(a) Requirement and inspection procedure shall be per MIL-P-60312. Two (2) samples shall be randomly selected from a quantity of trays for scheduled production for five (5) consecutive shifts. If one or more samples are defective, the quantity of trays represented by the samples shall be rejected. Inspection shall be accomplished within 24 hours of scheduled use of the trays.

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PAGE 10 (cont)

(b) Moisture content determination not required when this packing material is stored in an environmentally controlled area maintained at 40% Max Relative Humidity (R.H.) @ 72 degrees F for at least 24 hours prior to using."

PAGE 12

4.3.2.6, Add new Majors and notes as follows:

"101	Moisture content excessive for any packaging component (4 items)	(a)	5.1	Gage
102	Environmental conditions inadequate Gage	(b)	5.1	

(a) Components shall be inspected prior to use every hour during the production shift. If any sample exceeds the requirement, the inspection lot shall be rejected from further processing.

(b) Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing."

PAGE 16

* 4.4.3.2, Delete in its entirety and substitute:

"4.4.3.2 Weight of Propellant"

* Add paragraphs 4.4.3.2.1 and 4.4.3.2.2 as follows:

"4.4.3.2.1 Prior to assembly of cap (See dwg 9293422). The propellant charge shall be volumetrically dispensed (or tare-weighed) in a container, transferred to the ignition cartridge and then check-weighed 100 percent. The check-weighing shall be as specified in paragraph 4.5.4.1. The volumetric dispensing (or initial weighing) and check-weighing shall be performed by different operators on different scales. Weight limits for weighing shall be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective and removed from the lot.

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4.4.3.2.2 After assembly of cap (See dwg 9293422). Critical defect - more than 10 grains above assessed maximum weight: Critical defect - more than 10 grains below assessed minimum weight: Major defect - 1 to 10 grains above assessed maximum weigh: Major defect - 1 to 10 grains below assessed minimum weight. 315 sample ignition cartridge assemblies shall be randomly selected from each lot for this test. If any sample exceeds the assessed maximum weight by more than 10 grains or if two or more samples exceed the assessed maximum weight by 1 to 10 grains, the lot shall be rejected. If any sample is under the assessed minimum weight by more than 10 grains or if two or more samples are under the assessed minimum weight by 1 to 10 grains, the lot shall be rejected. Testing shall be as specified in paragraph 4.5.4.2."

* 4.4.3.3, Delete in its entirety.

* 4.4.3.3.1, Delete in its entirety.

* 4.4.3.3.2, Delete in its entirety.

PAGE 18

4.4.3.5, Delete paragraph in its entirety and substitute:

"4.4.3.5 Leak test (see 3.5). Sixty (60) ignition cartridges from each day's production shall be leak tested as specified in 4.5.3. If more than 3 failures occur, that day's production shall be 100% screened."

PAGE 19

4.4.3.8, Delete paragraph in its entirety.

4.5.3, Add at the beginning of the paragraph as follows:

"After the ignition cartridges have been assembled and the RTV is applied, the ignition cartridges will be kept in the controlled environmental storage for two (2) hours minimum prior to performing the leak test."

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* 4.5.4, Delete in its entirety and substitute:

"4.5.4 Weight of Propellant."

* Add paragraphs 4.5.4.1 and 4.5.4.2 as follows:

"4.5.4.1 Prior to assembly of cap. The ignition cartridge subassembly shall be weighed before and after the propellant is loaded. The weight of the propellant charge shall be determined by subtracting the tare weight of the ignition cartridge subassembly from the gross weight (after loading). The contractor may elect to perform the 100 percent check-weighing operation with automated equipment and mandatory process controls. In such cases, process controls plans in accordance with data item description DI-P-1604 must be submitted to the technical agency for approval of the automated operation.

4.5.4.2 After assembly of cap. The propellant shall be removed from each ignition cartridge sample, placed on the pan of a precision or analytic balance and weighed. Alternatively, the ignition cartridge assembly may be weighed and reweighed after removal of the propellant: The weight of the propellant charge shall be determined by subtracting the weight of the empty ignition cartridge from the loaded (gross) weight. The weight of the propellant in each sample shall be determined to the nearest 0.01 grain."

* Add alternate procedure as follows:

"Alternate Procedure - The ignition cartridge loading assembly shall be weighed and the net weight of propellant determined by subtracting parts tare weight from the gross weight."

4.5.7, Delete paragraph in its entirety.

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